

STM32WL WIRELESS SERIES



The world's first SoC enabling
LoRa®, (G)FSK, (G)MSK, BPSK



STM32WL series: the STM32 pioneer in Sub-GHz wireless connectivity.
Easy-to-use, reliable and tailored for a wide range of RF applications.

The STM32WL series supports multi-modulation and delivers outstanding ultra-low-power performance.

This makes it an ideal choice for LPWAN and IoT developments.

Embedding an Arm® Cortex®-M4 core, an optional Cortex®-M0+ and a Sub-GHz transceiver in one single chip, the STM32WL series ensures wireless application flexibility with LoRaWAN® and is compatible with Sigfox, W-MBUS or many other proprietary protocols, in a fully open way.

WIRELESS CONNECTIVITY

Designed for Sub-GHz LPWANs, the STM32WL series comes with a LoRaWAN® stack and Sigfox stack, available free of charge on st.com in STM32CubeWL.

Thanks to a deep integration, the innovative and open architecture is optimized for flexible resource use, power management and helps lower BOM cost while offering a better user experience.

Developed using the same technology as the one implemented in our ultra-low-power STM32L4 microcontrollers, the STM32WL series, with its integrated Sub-GHz transceiver,

provides similar digital and analog peripherals for basic or complex application use cases requiring an extended battery life and a long RF range.

SYSTEM PERIPHERALS

- Modulations: LoRa®, (G)FSK, (G)MSK, BPSK
- Linear Frequency Range: 150 to 960 MHz
- Dual-power Outputs: up to 22 dBm and up to 15 dBm (Embedded PAs)
- Up to 43 GPIOs
- Embedded SMPS and LDO
- Multiple low-power modes to maximize battery life

Standard protocol



- Available from STM32CubeWL
- Available from stackforce

Hardware tools

This STM32WL Nucleo pack is the most cost-effective way to quickly get started developing STM32WL-based prototypes.

Order codes

NUCLEO-WL55/C1
(868/915/923 MHz)
NUCLEO-WL55/C2
(433/470 MHz)



STM32WL portfolio

Flash memory / RAM size (bytes)



Notes:

- STM32WL55/56 all modulations available
- STM32WL54/54 all modulations available except LoRa

Legend:

- Dual-core
- Single-core

MULTIMEDIA

- Online Training WL: www.st.com/stm32wl-online-training
- Wireless YouTube Playlist: www.st.com/long-range-stm32wl
- STM32WL blog articles: www.blog.st.com

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